APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Quadratic Polynomial Factorization

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 9 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 90 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Solve the 15 standard factorization and middle-term splitting questions with discriminant analysis.

1. Executive Overview & Problem Context

A quadratic polynomial is a second-degree polynomial defined by P(x) = ax² + bx + c (a "` 0). Factoring factorizes the quadratic into linear binomial factors (px + q)(rx + s), identifying root solutions where the continuous parabolic curve intersects the horizontal Cartesian x-axis.

2. Core Analysis, Solutions & Systematic Working

- **Method of Middle-Term Splitting:**

  To factor ax² + bx + c, identify two integers p and q such that p + q = b and pq = ac. Example: Factorize 6x² + 17x + 5. Here ac = 6 × 5 = 30, and b = 17. Integers 15 and 2 satisfy (15 × 2 = 30, 15 + 2 = 17). Rewrite: 6x² + 15x + 2x + 5 = 3x(2x + 5) + 1(2x + 5) = (2x + 5)(3x + 1).

- **Discriminant Analysis & Root Categorization:**

  The discriminant D = b² - 4ac dictates root characteristics: D > 0 yields two distinct real rational/irrational roots; D = 0 yields two coincident (repeated) real roots (perfect square trinomial); D < 0 yields two complex conjugate roots with zero real x-axis intersections.

- **The Quadratic Formula Derivation:**

  By completing the square on ax² + bx + c = 0: x² + (b/a)x = -c/a -> [x + b/(2a)]² = (b² - 4ac)/(4a²) -> x = [-b ± " (b² - 4ac)] / 2a. Verified across non-factorable irrational polynomials such as 2x² - 6x + 3 = 0.

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3. Detailed Data Matrix & Systematic Findings

Quadratic Equation	Coefficients (a,b,c)	Discriminant D = b^2-4ac	Binomial Factors	Real Roots (x)
$x^2 + 7x + 12 = 0$	a=1, b=7, c=12	49 - 48 = 1 (D > 0)	(x + 3)(x + 4)	x = -3, -4
$2x^2 - 7x + 3 = 0$	a=2, b=-7, c=3	49 - 24 = 25 (D > 0)	(2x - 1)(x - 3)	x = 1/2, 3
$4x^2 - 12x + 9 = 0$	a=4, b=-12, c=9	144 - 144 = 0 (D = 0)	$(2x - 3)^2$	x = 3/2 (Repeated)
$3x^2 + 5x - 2 = 0$	a=3, b=5, c=-2	25 + 24 = 49 (D > 0)	(3x - 1)(x + 2)	x = 1/3, -2
$x^2 - 6x + 13 = 0$	a=1, b=-6, c=13	36 - 52 = -16 (D < 0)	Non-real linear factors	x = 3 ± 2i (Complex)

4. Student Conclusions & Reflective Synthesis

Quadratic algebra underpins kinematic projectile modeling, electrical impedance calculations, and architectural parabolic arch optimization in structural engineering.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Very good effort. Clear and organized factoring steps. Discriminant classification and graphical roots are thoroughly explained."

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